

Seguretat i Secret en la Codificació de la Informació

Enric Ventura

Departament de Matemàtiques

Graus en Enginyeria

Escola Politècnica Superior d'Enginyeria de Manresa

Universitat Politècnica de Catalunya

Curs 2022-2023

- 1 Modular Arithmetic Tools
- 2 Introduction to Cryptography
- 3 Introduction to Coding Theory

- 1 Modular Arithmetic Tools
- 2 Introduction to Cryptography
- 3 Introduction to Coding Theory

1.1. Rings and fields

1. Definition

A set R together with two operations $+: R \times R \rightarrow R$ and $\cdot: R \times R \rightarrow R$ is a *ring* if it satisfies the following properties:

- associatives: $\forall a, b, c \in R, \quad a + (b + c) = (a + b) + c, \quad a(bc) = (ab)c;$
- commutativity for $+$: $\forall a, b \in R, \quad a + b = b + a;$
- distributivity: $\forall a, b, c \in R, \quad a(b + c) = ab + ac;$
- neutral for $+$: $\exists e \in R \forall a \in R, \quad e + a = a;$
- oposed: $\forall a \in R \exists b \in R$ tal que $a + b = e;$
- neutral for $\cdot$: $\exists e \in R \forall a \in R, \quad ea = ae = a.$

If, additionally, $\cdot$ is commutative ($\forall a, b \in R, \quad ab = ba$) we say that R is a *commutative ring*.

2. Example

The ring of integers $\mathbb{Z}$. The rings $\mathbb{Q}, \mathbb{R}$. The matrix rings $M_n(\mathbb{Z}), M_n(\mathbb{Q}), M_n(\mathbb{R})$.

3. Observation

In any ring R , the neutral for $+$ is unique (and denoted 0), the neutral for $\cdot$ is unique (and denoted 1), and the opposed of a is unique (and denoted $-a$).

4. Observation

- (i) $\forall a \in R, \quad 0a = a0 = 0;$
- (ii) if $0 = 1$ then $|R| = 1$ (R is the trivial ring).

5. Definition

Let R be a ring. An element $a \in R$ is *invertible* if there exists $b \in R$ such that $ab = ba = 1$. In this case, b is *unique* and is denoted a^{-1} .

6. Definition

A ring R is a *field* if every element $a \neq 0$ is invertible.

7. Example

$\mathbb{Z}$ is not a field; $\mathbb{Q}$ and $\mathbb{R}$ are fields. Non of $M_n(\mathbb{Z})$, $M_n(\mathbb{Q})$, $M_n(\mathbb{R})$ is a field.

1.2. The ring of integers

8. Definition

Let $a, b \in \mathbb{Z}$. We say that a *divides* b , that b is *divisible by* a , and that b is *multiple of* a if $\exists c \in \mathbb{Z}$ such that $b = ac$. We denote this by $a|b$.

9. Observation

- (i) $a|b, b|c \Rightarrow a|c$;
- (ii) $a|b, a|c \Rightarrow a|(b+c), a|(b-c)$;
- (iii) $a|b \Rightarrow a|bc$;
- (iv) $a|bc \not\Rightarrow a|b$ or $a|c$;
- (v) $a|b, b|a \Rightarrow a = \pm b$.

10. Definition

Let $a, b \in \mathbb{Z}$, $a \neq 0$. The *greatest common divisor* of a and b is denoted $\text{mcd}(a, b)$.

11. Theorem (Euclidean division)

Given $a, b > 0$ integers, $\exists! q, r \in \mathbb{Z}$ such that $a = bq + r$, and $0 \leq r < b$.

12. Observation

$\text{mcd}(a, b) = \text{mcd}(a \pm b, b)$. Hence, $\text{mcd}(a, b) = \text{mcd}(b, r)$.

13. Explanation

Euclidean algorithm for computing the greatest common divisor.

14. Theorem (Bézout's identity)

Given $a, b > 0$, and $d = \text{mcd}(a, b)$, $\exists \alpha, \beta \in \mathbb{Z}$ such that $d = \alpha a + \beta b$.

15. Corollary

Let $a, b, d > 0$ be integers. Then,

$$d = \text{mcd}(a, b) \Leftrightarrow (1) d|a, \quad (2) d|b, \quad (3) c|a, c|b \Rightarrow c|d.$$

16. Definition

Two integers $a, b > 0$ are **coprime** if $\text{mcd}(a, b) = 1$ (or, equivalently, if $\exists \alpha, \beta \in \mathbb{Z}$, $\alpha a + \beta b = 1$).

17. Corollary

Let $a, b > 0$, $d = \text{mcd}(a, b)$, $a = da'$, $b = db'$. Then, a' and b' are coprime.

18. Corollary

If $a|bc$ and a and b are coprime, then $a|c$.

19. Definition

Let $a, b \geq 0$ be integers. The (non-negative) *least common multiple* of a and b is denoted $\text{mcm}(a, b)$.

20. Proposition

Let $a, b > 0$ be integers. Then,

(i) $\text{mcd}(a, b) \text{mcm}(a, b) = ab$;

(ii) $m = \text{mcm}(a, b) \Leftrightarrow (1) a|m, (2) b|m, \text{ i } (3) a|c, b|c \Rightarrow m|c$.

21. Theorem (Diophantine equations)

The Diophantine equation $ax + by = c$ has an integer solution if and only if $\text{mcd}(a, b)|c$. Description of all solutions.

22. Example

1.3. Arithmetic with integers

23. Definition

An integer $p \in \mathbb{Z}$ is *prime* if $p \neq 0, \pm 1$ and its unique divisors are ± 1 and $\pm p$.

24. Observation

Let $p \in \mathbb{Z}$ be a prime. If $p|ab$ then either $p|a$ or $p|b$.

25. Theorem (Fundamental for arithmetic)

Let $a \in \mathbb{Z}$, $a \neq 0, \pm 1$. Then, there exist pairwise different primes $p_1, \dots, p_n$ and naturals $r_1, \dots, r_n \geq 1$ such that $a = p_1^{r_1} \cdots p_n^{r_n}$. Moreover, $p_1, \dots, p_n$ and $r_1, \dots, r_n$ are unique up to signum and order.

26. Theorem

In $\mathbb{Z}$ there are infinitely many primes.

27. Explanation

Looking up to the square root. Sieve of Eratosthenes.

28. Proposition

Let $0 < a = p_1^{r_1} \cdots p_n^{r_n}$ and $0 < b = p_1^{s_1} \cdots p_n^{s_n}$ be the decompositions of a and b (with $r_i, s_i \geq 0$). Then, $\text{mcd}(a, b) = p_1^{\min\{r_1, s_1\}} \cdots p_n^{\min\{r_n, s_n\}}$ and $\text{mcm}(a, b) = p_1^{\max\{r_1, s_1\}} \cdots p_n^{\max\{r_n, s_n\}}$.

1.4. Modular arithmetic

29. Definition

Let $a, b, m \in \mathbb{Z}$. $a - b$ is multiple of $m \Leftrightarrow a$ and b both have the same reminder when dividing by m . In this case, we say that they are *congruent modulo m* , denoted $a \equiv b \pmod{m}$.

30. Proposition

$\equiv$ is an equivalence relation:

- (i) reflexive: $a \equiv a \pmod{m}$;
- (ii) symmetric: $a \equiv b \pmod{m} \Rightarrow b \equiv a \pmod{m}$;
- (iii) transitive: $a \equiv b \pmod{m} \wedge b \equiv c \pmod{m} \Rightarrow a \equiv c \pmod{m}$.

31. Definition

Equivalence classes modulo m : $[a]_m = \{b \in \mathbb{Z} \mid a \equiv b \pmod{m}\}$.

32. Observation

$\mathbb{Z} = [0]_m \sqcup [1]_m \sqcup \dots \sqcup [m-1]_m$ is a partition of $\mathbb{Z}$ in m classes. The set of classes is denoted $\mathbb{Z}/m\mathbb{Z} = \{[0]_m, [1]_m, \dots, [m-1]_m\}$.

33. Observation

- (i) $a \equiv a' \pmod{m}, b \equiv b' \pmod{m} \Rightarrow a + b \equiv a' + b' \pmod{m};$
- (ii) $a \equiv a' \pmod{m}, b \equiv b' \pmod{m} \Rightarrow ab \equiv a'b' \pmod{m}.$

34. Theorem

The operations in $\mathbb{Z}/m\mathbb{Z}$ given by $[a]_m + [b]_m = [a + b]_m$ and $[a]_m[b]_m = [ab]_m$ are well defined and turn the set $\mathbb{Z}/m\mathbb{Z}$ into a commutative ring.

35. Proposition

In the ring $\mathbb{Z}/m\mathbb{Z}$ we have:

- (i) $[a]_m$ is invertible $\Leftrightarrow a$ and m are coprime;
- (ii) $\mathbb{Z}/m\mathbb{Z}$ is a field $\Leftrightarrow m$ is prime;
- (iii) if $[a]_m$ is not invertible $\Rightarrow \exists [b]_m \neq [0]_m$ such that $[a]_m[b]_m = [0]_m.$

36. Observation

In the ring $\mathbb{Z}/m\mathbb{Z},$

- (i) if $[a]_m$ is invertible, then $[a]_m[b]_m = [a]_m[c]_m \Rightarrow [b]_m = [c]_m;$
- (ii) if $[a]$ is not invertible, the above is not true.

37. Explanation

For matrices modulo m , i.e., in the ring $M_n(\mathbb{Z}/m\mathbb{Z})$, the notion of *determinant* works exactly as (and with the same properties as) in $M_n(\mathbb{Z})$. In particular, $\det(AB) = \det(A) \cdot \det(B)$.

38. Proposition

A matrix $A \in M_n(\mathbb{Z}/m\mathbb{Z})$ is invertible $\Leftrightarrow \det(A) \in \mathbb{Z}/m\mathbb{Z}$ is invertible. In this case, A^{-1} can be computed in the classical way.

39. Explanation

Over a field, $\mathbb{Z}/p\mathbb{Z}$, p prime, all the standard linear algebra works exactly equal (*vectors, linear dependence, systems of equations, Cramer rule, ranks, determinants, etc*). If p is not a primer, everything is more complicated.

1.5. The Chinese reminder theorem

40. Explanation

Origin of the problem: a text by Sun Tsu Suan-Ching, from s. IV.

41. Observation

Let $m_1, \dots, m_n > 1$ and $a_1, \dots, a_n$ be integers. If the system of congruences

$$\begin{cases} x \equiv a_1 & (\text{mod } m_1) \\ \dots & \dots \\ x \equiv a_n & (\text{mod } m_n) \end{cases} \quad (1)$$

has solution, then it is unique modulo $M = \text{mcm}(m_1, \dots, m_n)$.

42. Corollary

The homogeneous case, $a_1 = \dots = a_n = 0$.

43. Theorem (Chinese remainder theorem)

If $m_1, \dots, m_n$ are all coprime to each other ($\text{mcd}(m_i, m_j) = 1, \forall i \neq j$), then the system of congruences (1) has a solution.

44. Example

45. Theorem

Let $m_1, \dots, m_n > 1$ be integers, and let $M = \text{mcm}(m_1, \dots, m_n)$. The following are equivalent:

- (a) the m_i 's are coprime to each other ($\text{mcd}(m_i, m_j) = 1, \forall i \neq j$);
- (b) $M = m_1 \cdots m_n$;
- (c) the following map is bijective:
$$\begin{aligned} f: \mathbb{Z}/M\mathbb{Z} &\rightarrow \mathbb{Z}/m_1\mathbb{Z} \times \cdots \times \mathbb{Z}/m_n\mathbb{Z}. \\ [x]_M &\mapsto ([x]_{m_1}, \dots, [x]_{m_n}) \end{aligned}$$

1.6. Big powers modulo m

46. Explanation

How to compute powers quickly, in a ring.

47. Theorem (Fermat's little theorem)

If p is prime and $p \nmid a$, then $a^{p-1} \equiv 1 \pmod{p}$.

48. Explanation

Caution!! $r \equiv s \pmod{m} \not\Rightarrow a^r \equiv a^s \pmod{m}$.

49. Explanation

Fast algorithm to show that something is not a prime.

50. Explanation

Fermat's little theorem does not work if p is not a prime.

51. Definition

We define the *Euler phi function* as the map $\phi: \mathbb{N} \rightarrow \mathbb{N}$ given by $m \mapsto \phi(m) = |\{r \in \{1, \dots, m\} \mid \text{mcd}(r, m) = 1\}|$.

52. Example

$\phi(1) = 1$, $\phi(2) = 1$, $\phi(3) = 2$, $\phi(10) = |\{1, 3, 7, 9\}| = 4$.

53. Observation

The ring $\mathbb{Z}/m\mathbb{Z}$ contains, precisely, $\phi(m)$ invertible elements.

54. Theorem

The Euler phi function ϕ has the following properties:

- (i) for p a prime, $\phi(p) = p - 1$;
- (ii) if $m \geq 3$ then $\phi(m)$ is even;
- (iii) for p a prime, if $p|m$ then $\phi(pm) = p\phi(m)$;
- (iv) for p a prime, if $p \nmid m$ then $\phi(pm) = (p - 1)\phi(m)$;
- (v) if $m = p_1 \cdots p_k$ is a product of pairwise different primes, then
$$\phi(m) = (p_1 - 1) \cdots (p_k - 1);$$
- (vi) if $m = p_1^{\alpha_1} p_2^{\alpha_2} \cdots p_k^{\alpha_k}$ is the decomposition of m in powers of different primes, then
$$\phi(m) = m \left(1 - \frac{1}{p_1}\right) \cdots \left(1 - \frac{1}{p_k}\right);$$
- (vii) if $\text{mcd}(m, n) = 1$ then $\phi(mn) = \phi(m)\phi(n)$.

55. Theorem (Euler)

Let $m > 1$ and a be two coprime integers. Then, $a^{\phi(m)} \equiv 1 \pmod{m}$.

56. Example

Calculation of big powers modulo m .